

Mechanical Design & Structural Simulation Report

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Software Used:

1. **SolidWorks**

2. **ANSYS**

3.

TASK 2: Structural Simulation (FEA)

Objective

To perform numerical grid discretization and structural static evaluation of the L-Bracket under custom boundary constraints and structural stress vectors.

Software Framework

ANSYS Workbench 2026 R1 (Static Structural Module)

Material Mechanics (Structural Steel)

Young's Modulus (E): 200GPa

Poisson's Ratio: 0.3

Tensile Yield Strength: 250MPa

Boundary Conditions and Environmental Setups

Fixed Support: Applied directly to the internal cylindrical faces of the bottom base mounting holes (0.30 mm) to restrict all 6 degrees of freedom ($U_x, U_y, U_z = 0$), mapping a rigid physical bolted connection.

Applied Structural Load (Force): A directional structural force vector was introduced on the functional face of the upper hole interface. The loading component was scaled along the gravity vector (Negative Y-Axis Component = -1N).

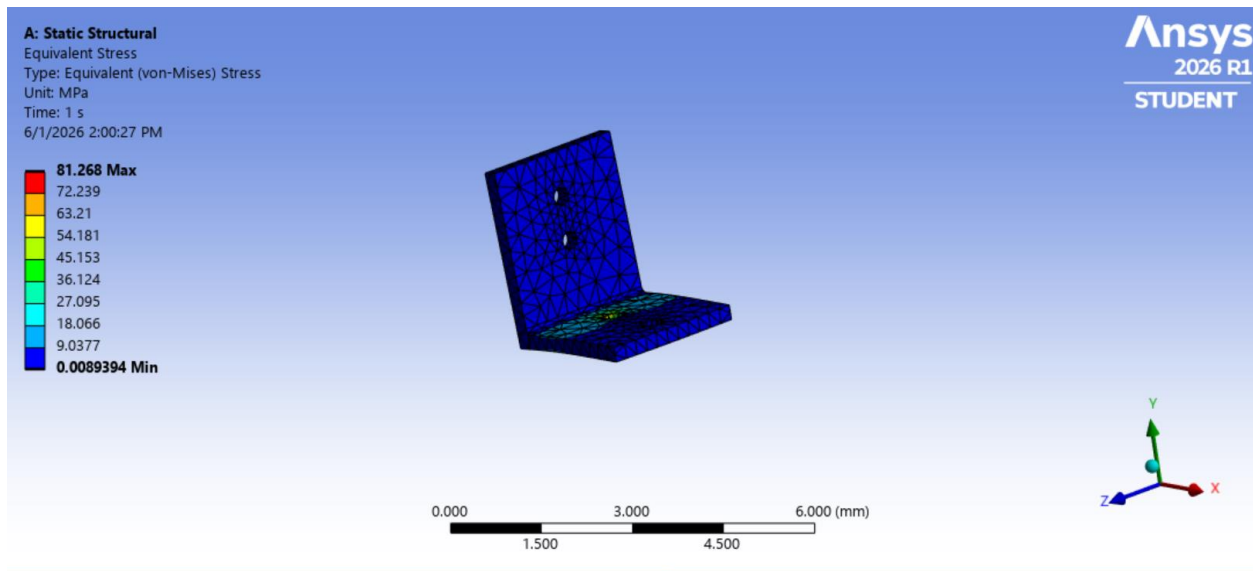
Discretization (Mesh): Automatically calculated adaptive solid tetrahedral elements were mapped across the thin-walled geometry to achieve high mathematical convergence around the high-stress geometric corners.

FEA Convergence Results and Verification:

1. Equivalent (von-Mises) Stress Distribution

Maximum Output	81.268MPa
Minimum Output	0.0089394MPa

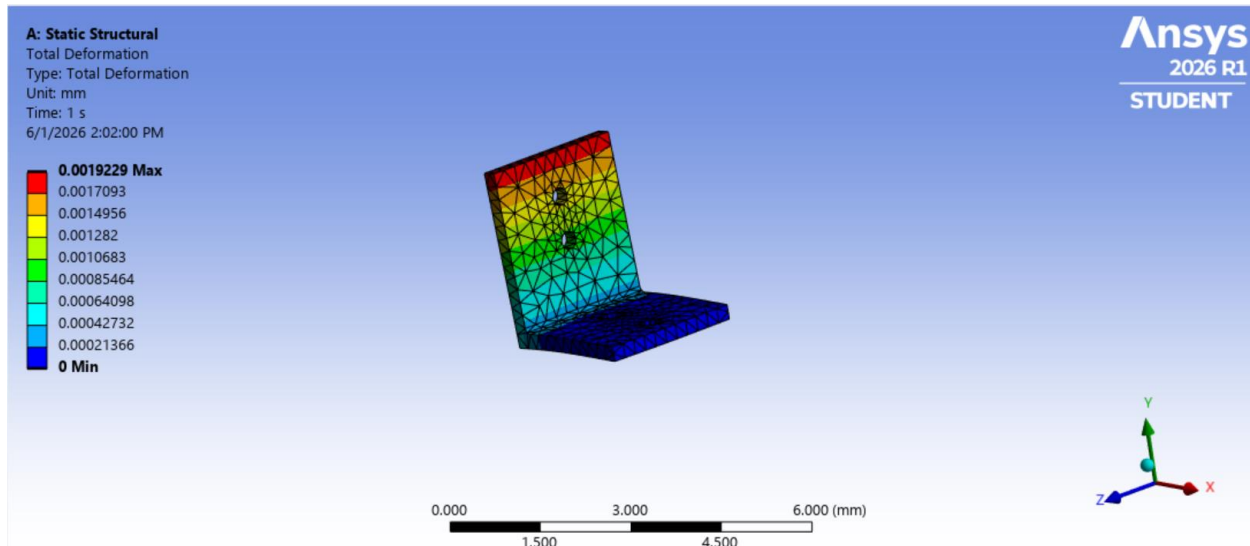
Engineering Evaluation: The continuous color contour map reveals that the maximum stress gradient is clustered closely around the inner radius bend zone and near the boundary fixed anchors. This response is highly compliant with standard beam bending mechanics theory. The maximum stress does not cross the material yield margin.



2. Total Deformation

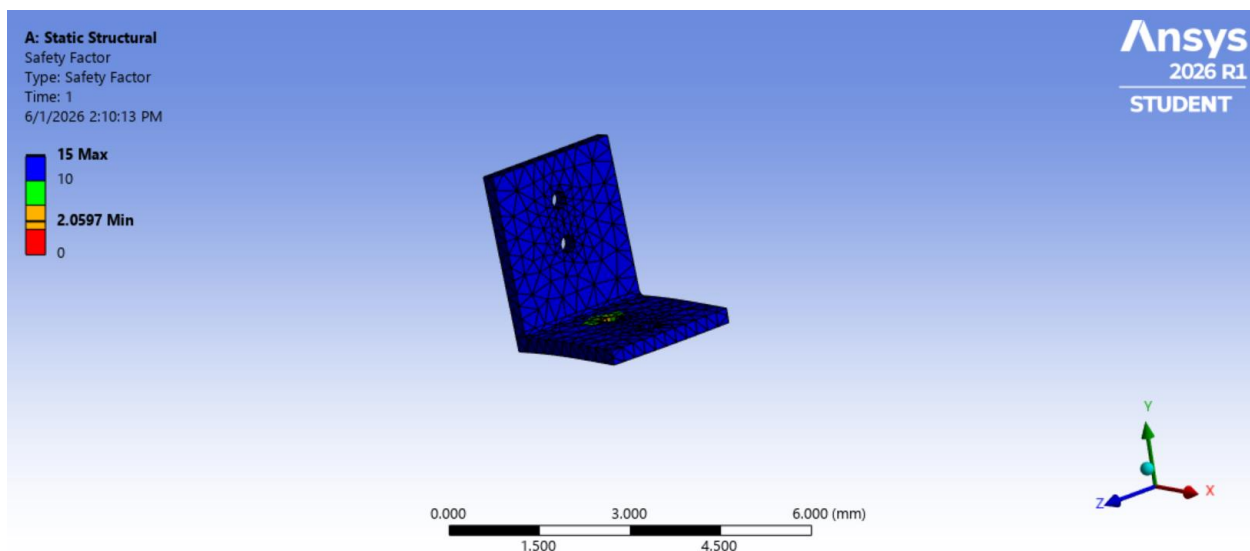
Maximum Output	0.0019229 mm
Minimum Output	0 mm (At fixed support faces)

Engineering Evaluation: The displacement field transitions smoothly up the vertical profile. The highest deflection occurs at the absolute topmost perimeter tip of the bracket where the moment arm is maximum. A maximum deflection of less than 2 indicates structural rigidity.



3. Factor of Safety (FOS)

Minimum Factor of Safety: 2.0597 (Locally concentrated at the inner high-stress root)



Final Engineering Observations & Conclusion

1. Design Margin: The calculated minimum Factor of Safety (2.0597) is greater than 1.0, proving the L-bracket can easily sustain the target load conditions without experiencing localized plastic deformation.

2. Stress Compatibility: The structural stress peak (81.268MPa) stays safely clear of the structural steel failure initiation line (250MPa).

3. Conclusion: The CAD framework from SolidWorks translates perfectly into the FEA solver matrix. The component design is mathematically declared structurally sound, optimized, safe and ready for validation.